Experiment 03_2 — Text Model Optimisation (TF-IDF & LinearSVC)

Goal: Find the best settings for the classic TF-IDF text model via cross-validation.

1. **Grid Search (3-Fold CV)**: Systematic evaluation of binary weighting and SVM regularisation (C) on the training split.
2. **Classifier Comparison**: Benchmark `LinearSVC` against `LogisticRegression` (SGD) and `ComplementNB`.
3. **Validation Verification**: Evaluate the final optimised model once on the shared 16984 validation products.

1. Load the data

```
Train:      67932
Validation: 16984
```

2. Grid search on LinearSVC

The 03 notebook showed that using all features beats a 50000 limit by 1.2 points (CV on train). So we keep all features.

We test two settings together:

- Binary TF-IDF: count only whether a word appears, not how often (yes or no)
- C: how closely the SVM fits the training data. A small C gives a simpler model.

That gives 8 combinations, each scored with 3-fold CV on train.

```
Grid search time: 1.9 min
```

Binary	C	CV weighted F1	CV spread
True	1.0000	0.8284	0.0026
True	2.0000	0.8282	0.0020
True	0.5000	0.8263	0.0030
False	1.0000	0.8235	0.0025
False	2.0000	0.8231	0.0024
False	0.5000	0.8223	0.0025
True	0.1000	0.8031	0.0013
False	0.1000	0.7999	0.0017

```
Binary gain:  +0.0050
CV spread:    0.0030
Use binary:   True
Best C:       1.0
```

Analysis

- C = 1 gives the best score. C = 0.5 and 2 are almost the same. C = 0.1 is about 2.5 points lower.
- Binary scores 0.8284 against 0.8235 without it. The gain (0.5 points) is larger than the CV spread (0.3), so we keep binary TF-IDF.
- Chosen settings: all features, binary TF-IDF, C = 1.

3. Other models

We keep the chosen TF-IDF settings and only change the classifier:

- Complement Naive Bayes: a fast and simple text model.
- Logistic Regression (trained with SGD): a linear model like the SVM that also gives probabilities.

Each model is scored with the same 3-fold CV on train.

Model	CV weighted F1	CV spread	Time (min)
LinearSVC (chosen)	0.8284	0.0026	0.3041
Logistic Regression (SGD)	0.8084	0.0015	0.1933
Complement Naive Bayes	0.7600	0.0003	0.1199

Analysis

- LinearSVC has the best score (0.8284).
- Logistic Regression is 2 points lower (0.8084).
- Complement Naive Bayes is 7 points lower (0.7600).
- We keep LinearSVC.

4. Final score on validation

We train the baseline and the best model on the full training set, and score both once on validation.

Model	Weighted F1	Macro F1
Baseline (03 notebook)	0.8313	0.8192
Optimised	0.8436	0.8330

Analysis

- The optimised model reaches 0.8436 weighted F1 on validation, against 0.8313 for the baseline (+1.2 points).
- Macro F1 goes from 0.8192 to 0.8330.
- The gain comes from using all features. Binary TF-IDF and C = 1 were already used in the baseline.

Conclusion

Final text model: TF-IDF on words and word pairs (all features, binary) with LinearSVC (C = 1). Weighted F1 on validation: 0.8436.